MIL-F-51206(MU)
3 September 1970
(see section 6)

MILITARY SPECIFICATION

FILTER, COTTON

1. SCOPE

1*1 Scope This specification covers two forms of fines cotton filter.

1.2 Classification. The cotton shall be of the following forms:

Form 1 - Roll
Form 2 - Disc

2. APPLICABLE DOCUMENTS

2.1 Government documents. The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

STANDARDS

FEDERAL

FED-STD-191 - Textile Test Methods.

MILITARY

MIL-STD-125 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

DRAWINGS

US ARMY MUNITIONS COMMAND

EDGEWOOD ARSENAL

DLB38-6-351-7 - Q112 Canister Air Resistance Indicator.
DLEB1255-8-1-0 - Q101 Water Repellency Indicator.

(Copies of specifications, standards drawings and publications required by suppliers in connection with specific procwement functions " should be obtained form the procuring activity or as directed by the contracting officer.)

FSC 4240

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated,. the issue in effect on date of invitation for bids or request for proposal' , shall apply.

UNIFORM CLASSIFICATION COMMITTEE

Uniform Freight Classification

(Application for copies of these ratings, rules and regulations should be addressed to Uniform Classification Committee, 202 Union Station, 516 West Jackson Boulevard, Chicago, Illinois 60606.)

AMERICAN TRUCKING ASSOCIATIONS, INC.

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Associations, Inc. , 1616 P Street, N.W., Washington. D.C. 20036.)

CODE OF FEDERAL REGULATIONS

49 CFR 171-179 - Department of Transportation Rules and Regulations for the Transportation of Explosives and Other Dangerous Articles.

(The Department of Transportation regulations are a part of the Code of Federal Regulations available from the Superintendent of Documents, Government Printing Office, Washington, D.C. 20402. Orders for the above publication should cite "49 CFR 171-179.")

39 REQUIREMENTS

3.1 Material. The material shall be coarse cotton fibers containing unshredded yarns. The unshredded yarns shall be evenly distributed throughout the cotton filter.

3.2 Water repellency. The surfaces of the cotton filter shall be bonded with a water-repellent resin. The treated material shall support a column of water 4 inches in height when tested as specified in 4.4.4.1.

3.3 Airflow resistance. The cotton filter shall have an airflow resistance of 0.6 + 0.2 millimeter (mm) of water per 100 square centimeters when tested–as specified in 4.4.4.2.

3.4 Mildew resistance. The cotton filter shall be treated with a mildew-resistant agent of sufficient strength to prevent the growth of more than a trace of mildew when tested as specified in 4.4.4.3.

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3.5 Thickness. When tested as specified in 4.4.4.4, each thickness measurement shall be between 0.019 and 0.043 inch. The average of all measurements shall be 0.031 + 0.008 inch.

3.6 Size.

3.6.1 Disc. When specified in the contract or order, (see 6.2) the cotton filter shall be died into discs conforming to the drawing specified.

3.6.2 Roll. Unless otherwise specified in the contract or order, (see 6.2) the cotton filter shall be supplied in rolls not less than 90 yards nor more than 110 yards in total length. Each roll shall contain not more than three pieces and each piece shall be not less than 20 feet in length. The rolls shall have a nominal width of 36 inches. No feather edge shall be permitted.

3.7 Water extracts. The water extracts of the treated cotton filter shall be neutral to litmus when tested as specified in 4.4.4.5.

3.8 Preproduction. Prior to the start of regular production a preproduction sample of cotton filter shall be produces in accordance with this specification for examination and tests (see 4.3).

3.9 Workmanship. The cotton filter shall contain no tears, rips, perforations, holes, dirt, grease, oil contamination, or other imperfect ions which may impair its serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.

4.1.1 Supplier's responsibility. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to specified requirements.

4.1.2 Objective evidence. The supplier shall provide objective evidence acceptable to the contracting officer that the requirements of 3.1 and section 5, for which specific inspection has not been provided in this specification, have been satisfied.

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4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) Preproduction inspection (see 4.3).
(b) Quality confommnce inspection (see 4.4).

4.3 Preproduction inspection.

4.3.1 Sample. One roll of cotton filter or '50 discs shall be produced using the same methods, materials, equipment, and processes as will be used during regular production.

4.3.2 Inspection procedure.

4.3.2.1 For examination. A 6-foot long sample, full width no less than 6 feet from the end of roll, or 25 discs, shall be examined for all requirements of the applicable Drawings and this specification.

4.3.2.2 For tests. A total of five specimens each for 4.4.4.1, 4.4.4.2 and 4.4.4.3 and three specimens each for 4.4.4.4 and 4.4.4.5 shall be taken randomly from 4.3.2.1 samples.

4.3.3 Acceptance/rejection criteria. The preproduction sample shall meet the examination and tests specified by 4.3.2.1 and 4.3.2.2 to be acceptable, The supplier shall obtain written approval from the contracting officer before proceeding with regular production.

4.4 Quality conformance inspection.

4.4,1 Lotting. A lot shall consist of the cotton filter produced by one manufacturer, during one work shift, from the same materials, and under .essentially the same manufacturing conditions.

4.4.2 sampling,

4.4.2.I For examination. Sampling for rolls or discs shall be conducted in accordance with MIL-STD-105. For the rolls, a 6-foot long, full width sample shall be taken no less than 6 **feet** from the end of the roll from the selected rolls.

4.4.2.2 Tests. A total of five specimens each for 4.4.4.1, 4.4.4.2 and 4.4.4.3 and three specimens each for 4.4.4.4 and 4.4.4.5 shall be taken randomly from 4.4.2.1 samples.

4.4.3 Inspection procedure.

4.4.3.1 For examination. Each sample (4.4.2.1) shall be examined in accordance with the classification of defects and MIL-STD-105.

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4.4.3.2 For tests. Specimens taken in accordance with 4.4.2.2 shall be tested as specified in 4.4.4. The acceptance number is zero

4.4.3.3 Classification of defects.

(a) Filter, cotton,

Categories	Defects
Critical:	none defined
Major	None defined
Minor:	AQL 2.5 percent defective
201	Unshredded yarns (3.1)
202	Disc diameter incorrect (Applicable Drawing)
	AQL 10.0 percent defective
203	Workmanship (3.9)

4.4.4 Tests.

4.4.4.1 Water repellency. Specimens shall be tested for water repellency with the Q101 Water Repellency Indicator (DLB125-8-1-0).

4.4.4.2 Airflow resistance. Specimens shall be tested for airflow resistance with the Q1l.2 Canister Air Resistance Indicator (DLB 38-6-351-7).

4.4*4.3 Mildew resistance. Specimens shall be tested for mildew resistance in accordance with method 5750 of FED-STD-191.

4.4.4.4 Thickness. Specimens shall be tested for thickness by means of a dial-type thickness gage having a 3-ounce presser foot 3/8-inch in diameter. Thickness determinations shall be made at five different points on each specimen.

4.4.4.5 Water extracts. Specimens shall be tested for water extracts as follows: cut 2-3/4 square inch from the roll or disc and submerge the specimens in individual beakers containing approximately 50 ml of boiling distilled water. Continue to boil for 2 minutes, then test with red and blue litmus paper.

5. PREPARATION FDR DELIVERY

5.1 Packaging.

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5.1.1 Level c. Unit rolls or discs shall be packaged to provide adequate protection against moisture, dust, deterioration, and damage from the supply source to the first receiving activity for immediate use. ,

5.2 packing.

5.2.1 Level c. Unit packages of rolls or discs, packaged as specified in 5.1.1, shall be packed in containers which comply with requirements of the Code of Federal Regulations, Department of Transportation Regulations, Uniform Freight or National Motor Freight Classification, Ratings, Rules and Regulations applicable to the mode of transportation. Containers shall be capable of stacking and supporting superimposed loads during shipment and adequate to assure safe delivery to destination without damage to contents and acceptable for shipment at the most favorable rate. In addition to the above information, the discs shall be nested in its shipping container.

5.3 Marking. In addition to any special marking required by the contractor or order the unit packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. This specification covers cotton filter from which discs can be died for use as a fines filter in protective mask canisters.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Wether roll or discs are required and drawing number if discs are to be supplied.
(c) Preproduction.

(1) Time allowed for supplier submission of samples for Government test and evaluation after award of contract.
(2) Name and address of test facility and shipping instructions when testing is performed by the Government.
(3) Time required for the Government to notify the supplier whether or not to proceed with production.

6.3 Supersession data. This specification includes the requirements of Edgewood Arsenal Purchase Description 197-54-483A dated 11 October 1954.

Custodian:

Army - MU

Preparing actitity:

Army- MU(EA)

project No. 4240-A434

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD. FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 68 WHICH MAY BE USED.

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Commanding Officer
Edgewood Arsenal
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Edgewood Arsenal, Maryland 21010

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